

1 Theory

Multiplexer

A multiplexer (MUX) is a combinational logic circuit that selects one of several input signals and forwards the selected input to a single output line. It operates based on the binary values of select lines. The basic purpose of a multiplexer is to allow multiple data sources to share a single output channel, thereby reducing the number of lines required in a circuit. A multiplexer can be thought of as a digital switch that connects one of many inputs to a single output. The selection of the input line is controlled by the values of the select lines, which determine which input is routed to the output.

Functionality

The functionality of a multiplexer can be described using the following components:

Input Lines (I): The number of input lines determines how many data signals can be transmitted. For an n-to-1 multiplexer, there are n input lines.

Select Lines (S): The select lines determine which input line is connected to the output. The number of select lines is determined by the equation $s = \log_2(n)$ where n is the number of input lines.

Output Line (O): There is a single output line that carries the selected input signal.

Types of Multiplexers

Multiplexers are classified based on the number of inputs they can handle.

4-to-1 Multiplexer: Has four input lines, two select lines, and one output line.

8-to-1 Multiplexer: Has eight input lines, three select lines, and one output line.

16-to-1 Multiplexer: Has sixteen input lines, four select lines, and one output line.

⋮

n-to-1 Multiplexer: Has n input lines, $\log_2(n)$ select lines, and one output line.

Advantages of Multiplexers

Reduced Wiring Complexity: Multiplexers minimize the number of connections needed in a circuit.

Increased Efficiency: They enable the selection of signals without the need for additional components.

Flexibility: Multiplexers can easily adapt to various input and output configurations.

Demultiplexer

A demultiplexer (DEMUX) is a combinational logic circuit that takes a single input signal and routes it to one of several output lines. It operates based on the binary values of select lines. The primary purpose of a demultiplexer is to distribute data from a single source to multiple destinations, thereby enabling effective data management in digital circuits. A demultiplexer can be considered a reverse multiplexer, where the selection of the output line is controlled by the values of the select lines, determining which output receives the input signal.

Functionality

The functionality of a demultiplexer can be described using the following components:

Input Lines (I): Input Line (I): The number of input lines is always one, which carries the data signal to be distributed.

Select Lines (S): The select lines determine which output line will receive the input signal. The number of select lines is determined by the equation $s = \log_2(n)$ where n is the number of input lines.

Output Line (O): There are multiple output lines, with only one output line activated at any given time, depending on the values of the select lines

Types of Multiplexers

Demultiplexers are classified based on the number of outputs they can handle:

1-to-4 Demultiplexer: Has one input line, two select lines, and four output lines.

1-to-8 Demultiplexer: Has one input lines, three select lines, and eight output lines.

1-to-16 Demultiplexer: Has one input lines, four select lines, and sixteen output lines.

⋮

1-to-n Demultiplexer: Has one input lines, $\log_2(n)$ select lines, and n output lines.

Advantages of Multiplexers

Efficient Data Distribution: Demultiplexers allow for effective routing of data from a single source to multiple destinations.

Reduced Complexity in Circuit Design: By enabling the distribution of signals without the need for multiple input lines, demultiplexers simplify the overall design of digital circuits.

Increased Control: Demultiplexers provide enhanced control over data flow in systems, facilitating easier management of signals in various applications.

2 Implementation and Observation

2.1 4-1 MUX

VHDL code

```
--mux 4_1
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity mux_4_1 is
    Port ( i : in STD_LOGIC_VECTOR (3 downto 0);
          s : in STD_LOGIC_VECTOR (1 downto 0);
          o : out STD_LOGIC);
end mux_4_1;

architecture Behavioral of mux_4_1 is
begin

    process(s, i)
    begin
        if (s = "00") then
            o <= i(0);
        elsif (s = "01") then
            o <= i(1);
        elsif (s = "10") then
            o <= i(2);
        elsif(s = "11") then
            o <= i(3);
        else
            o <= '0';
        end if;
    end process;

end Behavioral;
```

RTL Schematic Diagram

RTL design is used to describe the behavior of the 4-1 MUX as shown in figure 1. It shows the process of how the input and selection lines associate with each other to provide desired output.

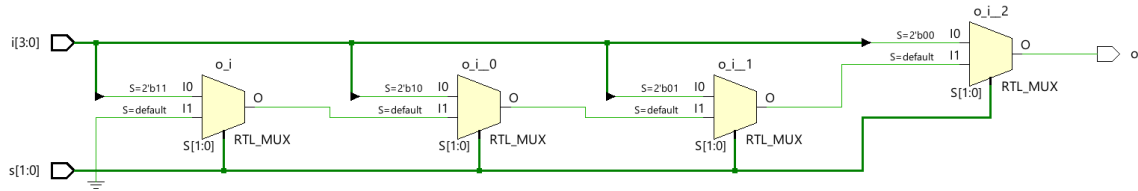


Figure 1: RTL Schematic Diagram of 4-1 MUX

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the 4-1 MUX circuit as shown in figure 2. It involves converting a VHDL code into a gate-level representation through synthesis, generating a netlist that optimizes for factors like speed and power. The resulting schematic visually represents the gates and interconnections, allowing engineers to verify and optimize the circuit's design for efficiency and functionality.

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table of 4-1 MUX as shown in figure 3. In the case of a 4-to-1 MUX, the test ensures that the behavior of the VHDL program and the corresponding schematic diagram match the expected behavior as defined by the truth table. The truth table outlines the relationship between the inputs, select lines, and output for all possible combinations.

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 4.

Power Estimation

Figure 5 shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a 4-1 MUX circuit such as information as junction temperature, thermal margin, confidence level and ambient temperature.

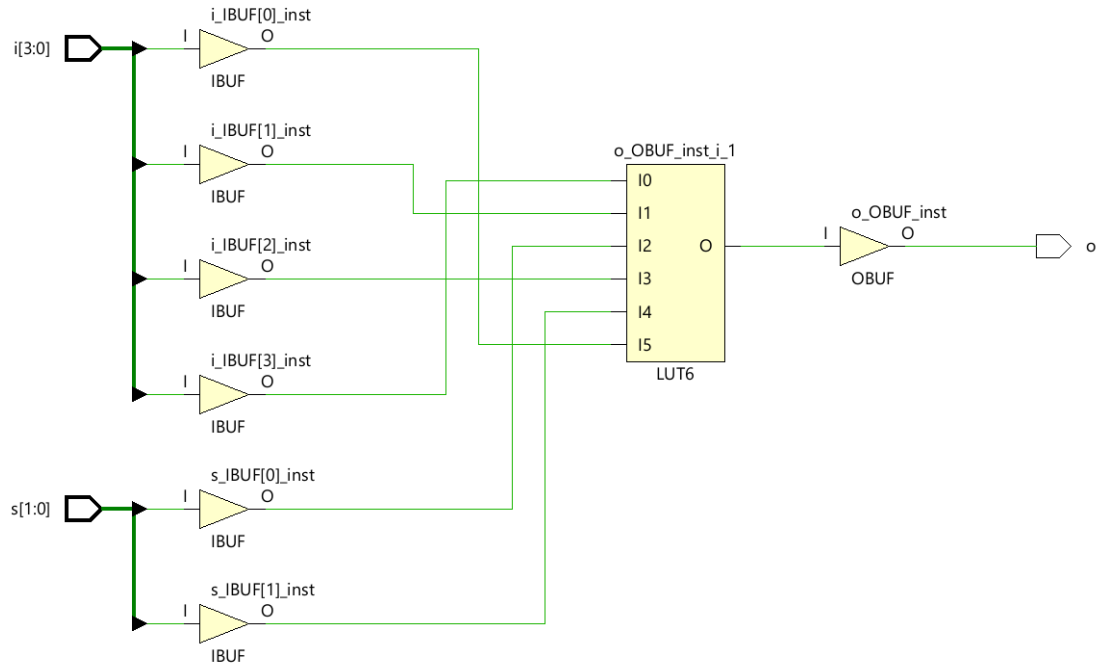


Figure 2: Synthesis Schematic Diagram of 4-1 MUX

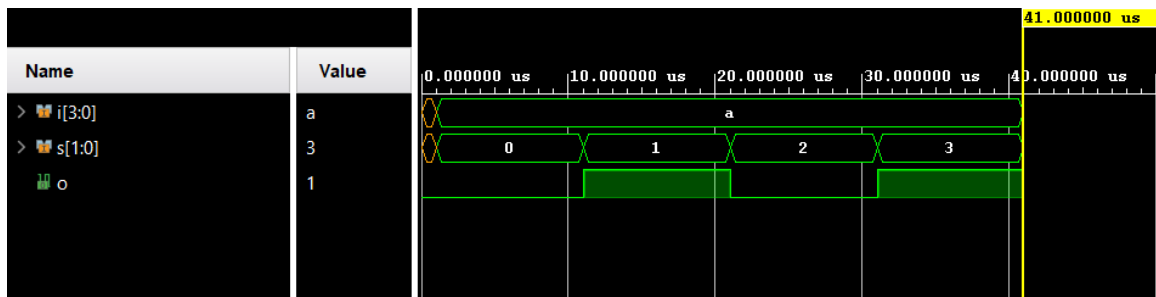


Figure 3: Timing Diagram of 4-1 MUX

2.2 8-1 MUX

VHDL code

```
--mux 8_1
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity mux_8_1 is
    Port ( i : in STD_LOGIC_VECTOR (7 downto 0);
          s : in STD_LOGIC_VECTOR (2 downto 0);
          o : out STD_LOGIC);
end mux_8_1;
```

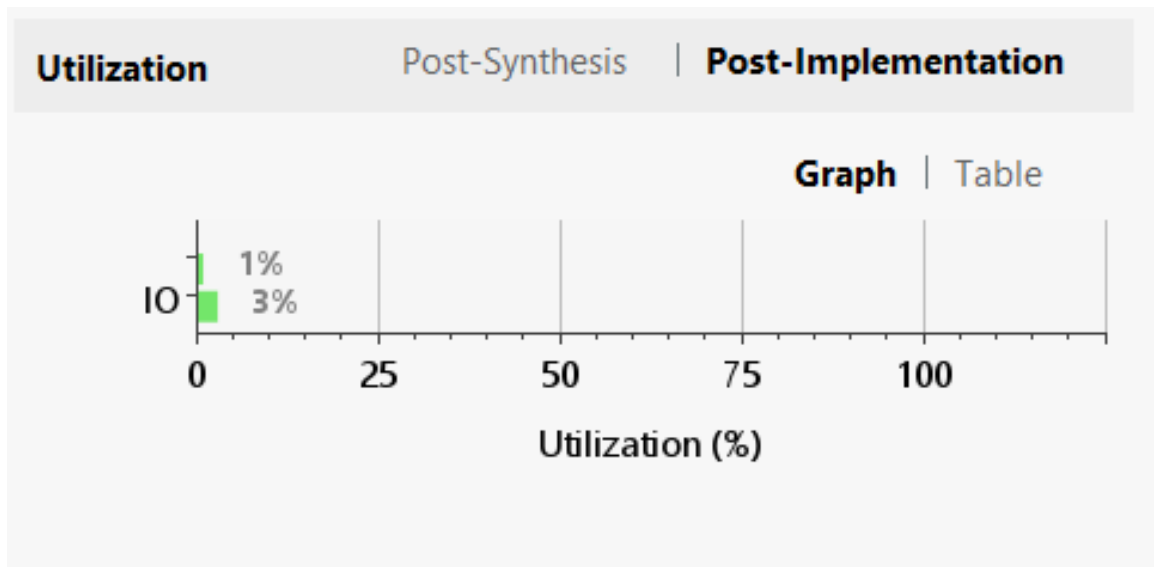


Figure 4: Device Utilization of 4-1 MUX

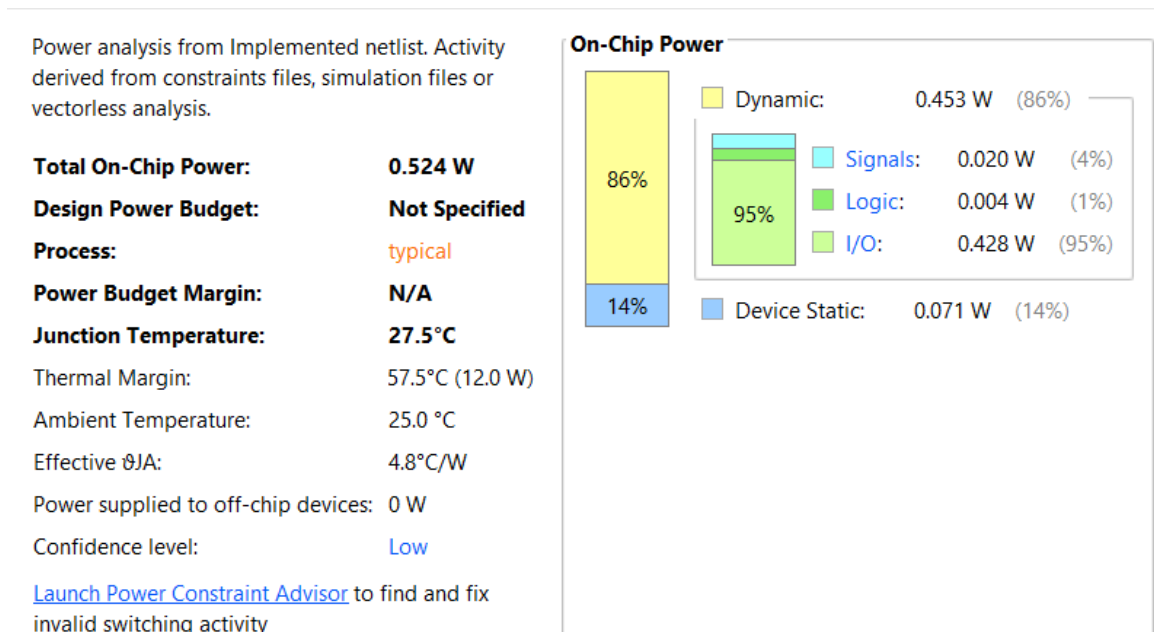


Figure 5: Power Estimation of 4-1 MUX

```
architecture Behavioral of mux_8_1 is
begin
```

```
    process(s, i)
    begin
        if (s = "000") then
            o <= i(0);
        elsif (s = "001") then
            o <= i(1);
        elsif (s = "010") then
            o <= i(2);
        elsif (s = "011") then
            o <= i(3);
        elsif (s = "100") then
            o <= i(4);
        elsif (s = "101") then
            o <= i(5);
        elsif (s = "110") then
            o <= i(6);
        elsif (s = "111") then
            o <= i(7);
        else
            o <= '0';
        end if;
    end process;
```

```
end Behavioral;
```

RTL Schematic Diagram

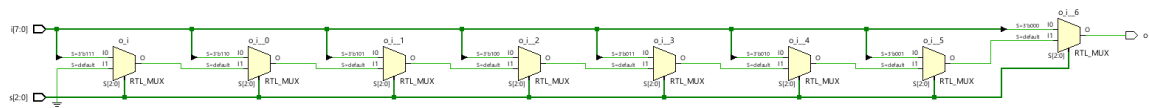
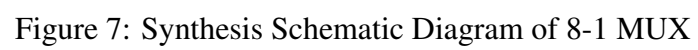


Figure 6: RTL Schematic Diagram of 8-1 MUX

RTL design is used to describe the behavior of the 8-1 MUX as shown in figure 6. It shows the process of how the input and selection lines associate with each other to provide desired output.



Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the 8-1 MUX circuit as shown in figure 7. It involves converting a VHDL code into a gate-level representation through synthesis, generating a netlist that optimizes for factors like speed and power. The resulting schematic visually represents the gates and interconnections, allowing engineers to verify and optimize the circuit's design for efficiency and functionality.

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table of 8-1 MUX as shown in figure 8. In the case of a 8-1 MUX, the test ensures that the behavior of the VHDL program and the corresponding schematic diagram match the expected behavior as defined by the truth table. The truth table outlines the relationship between the inputs, select lines, and output for all possible combinations.

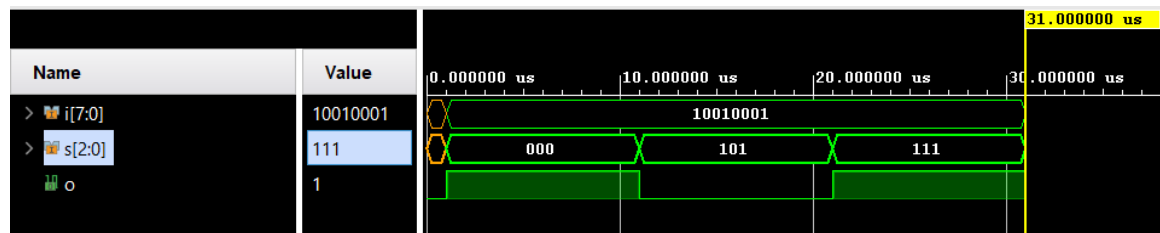


Figure 8: Timing Diagram of 8-1 MUX

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 9.

Power Estimation

Figure 10 shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a 8-1 MUX circuit such as information as junction temperature, thermal margin, confidence level and ambient temperature.

2.3 16-1 MUX

VHDL code

```
library IEEE;  
use IEEE.STD_LOGIC_1164.ALL;
```

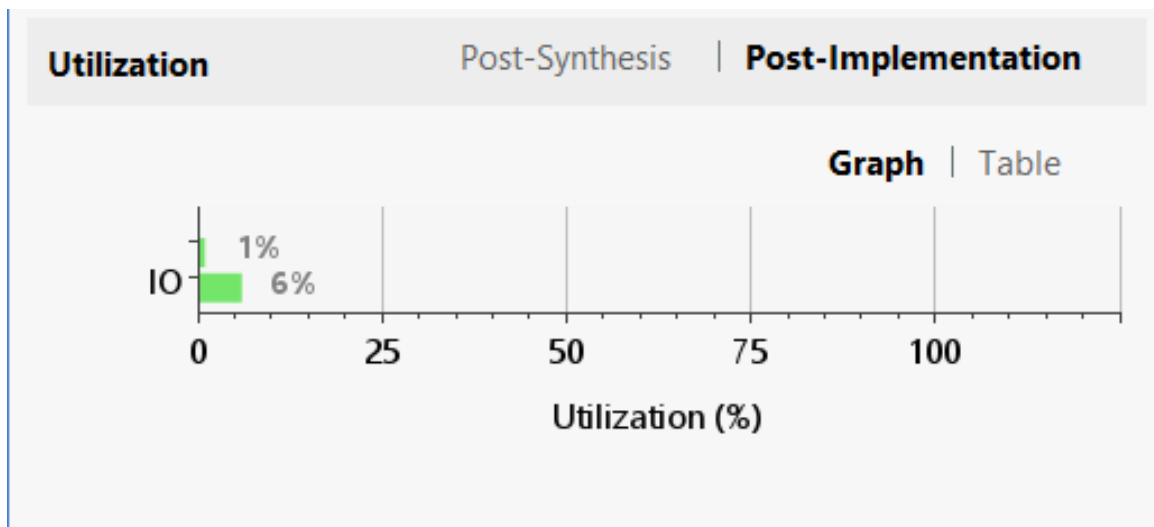


Figure 9: Device Utilization of 8-1 MUX

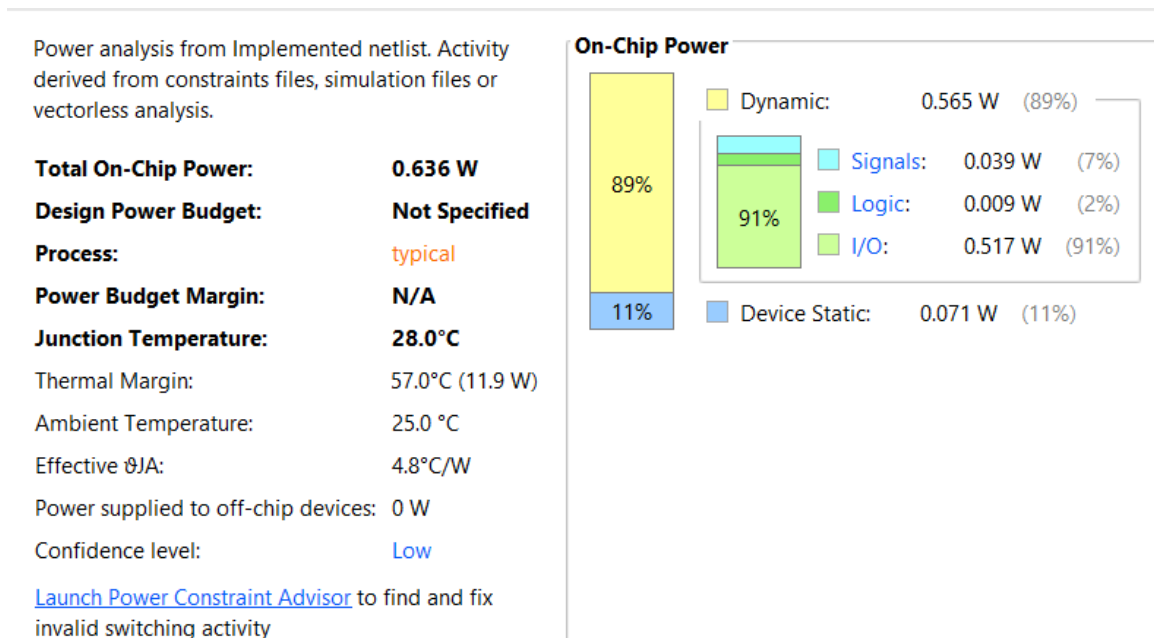


Figure 10: Power Estimation of 8-1 MUX

```

entity mux_16_1 is
    Port ( i : in STD_LOGIC_VECTOR (15 downto 0);
          s : in STD_LOGIC_VECTOR (3 downto 0);
          o : out STD_LOGIC);
end mux_16_1;

```

```

architecture Behavioral of mux_16_1 is
begin

```

```

    process(s, i)
    begin
        if (s = "0000") then
            o <= i(0);
        elsif (s = "0001") then
            o <= i(1);
        elsif (s = "0010") then
            o <= i(2);
        elsif (s = "0011") then
            o <= i(3);
        elsif (s = "0100") then
            o <= i(4);
        elsif (s = "0101") then
            o <= i(5);
        elsif (s = "0110") then
            o <= i(6);
        elsif (s = "0111") then
            o <= i(7);

            elsif (s = "1000") then
                o <= i(8);
            elsif (s = "1001") then
                o <= i(9);
            elsif (s = "1010") then
                o <= i(10);
            elsif (s = "1011") then
                o <= i(11);
            elsif (s = "1100") then
                o <= i(12);
            elsif (s = "1101") then
                o <= i(13);
            elsif (s = "1110") then
                o <= i(14);
            elsif (s = "1111") then
                o <= i(15);

```

```

        else
            o <= '0';
        end if;
    end process;

end Behavioral;

```

RTL Schematic Diagram

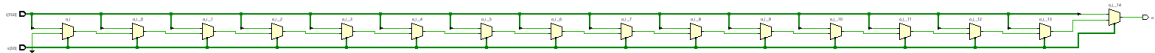


Figure 11: RTL Schematic Diagram of 16-1 MUX

RTL design is used to describe the behavior of the 16-1 MUX as shown in figure 11. It shows the process of how the input and selection lines associate with each other to provide desired output.

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the 16-1 MUX circuit as shown in figure 12. It involves converting a VHDL code into a gate-level representation through synthesis, generating a netlist that optimizes for factors like speed and power. The resulting schematic visually represents the gates and interconnections, allowing engineers to verify and optimize the circuit's design for efficiency and functionality.

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table of 16-1 MUX as shown in figure 13. In the case of a 16-1 MUX, the test ensures that the behavior of the VHDL program and the corresponding schematic diagram match the expected behavior as defined by the truth table. The truth table outlines the relationship between the inputs, select lines, and output for all possible combinations.

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 14.

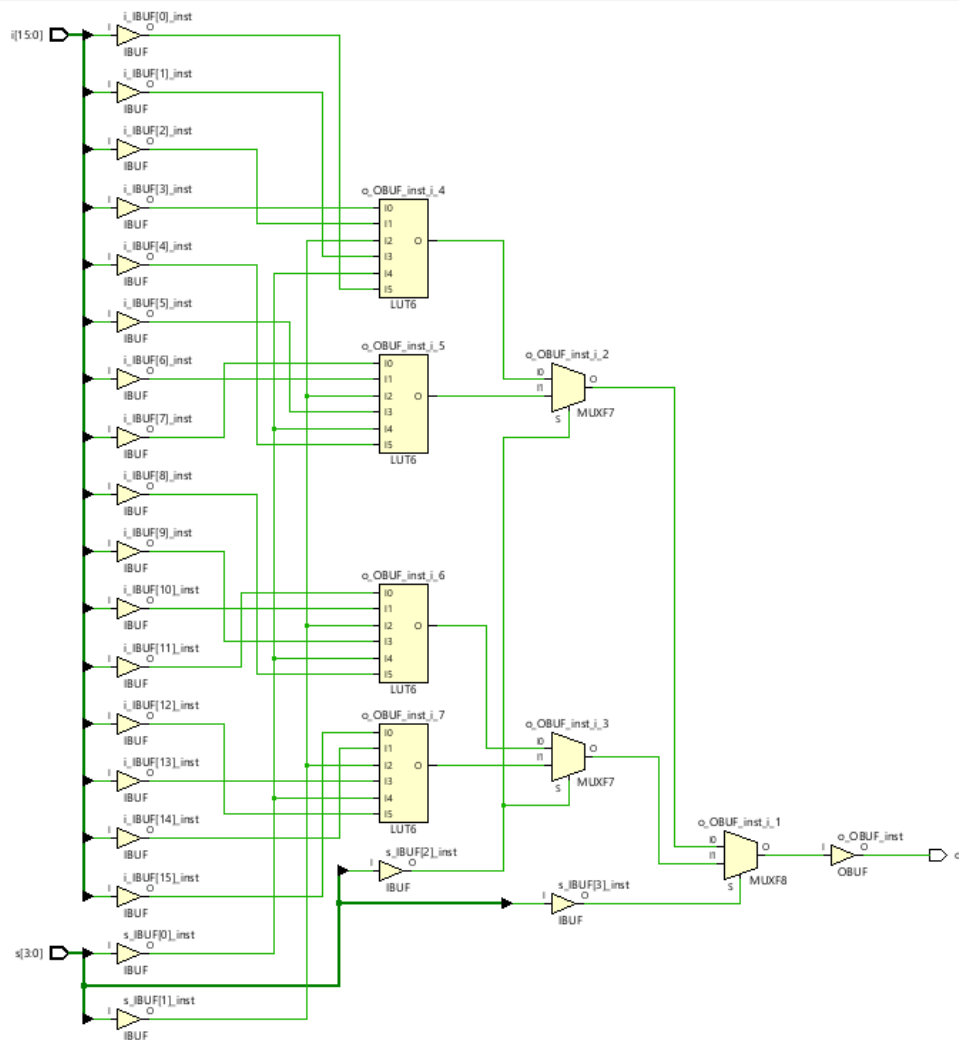


Figure 12: Synthesis Schematic Diagram of 16-1 MUX

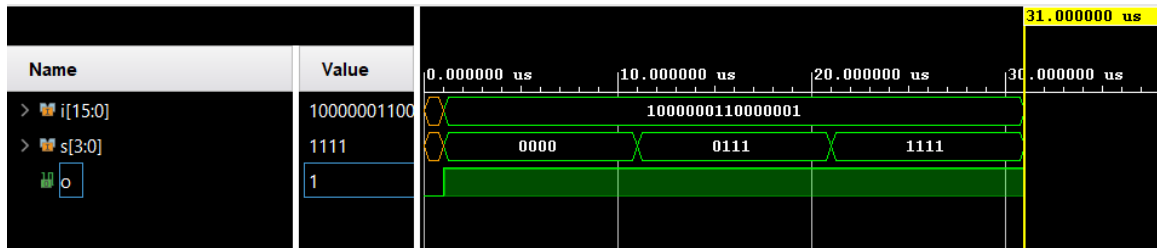


Figure 13: Timing Diagram of 16-1 MUX

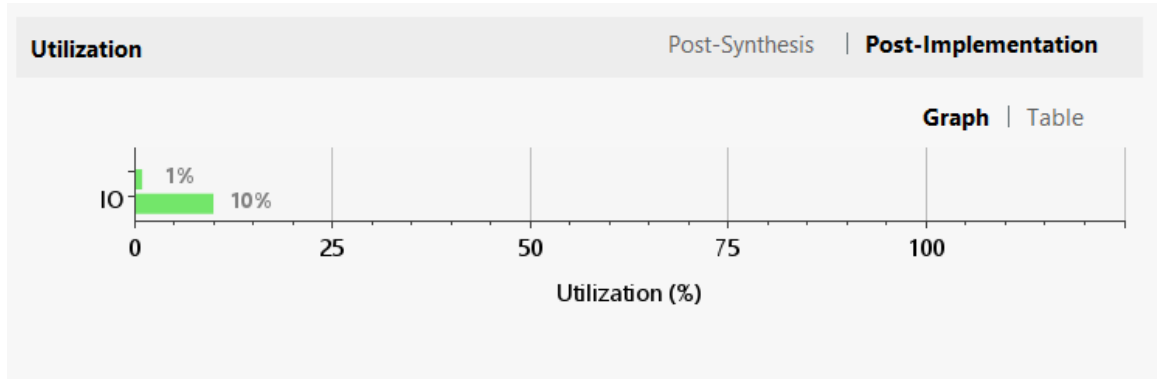


Figure 14: Device Utilization of 16-1 MUX

Power Estimation

Figure 15 shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a 16-1 MUX circuit such as information as junction temperature, thermal margin, confidence level and ambient temperature.

2.4 1-4 DEMUX

VHDL code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity demux_1_4 is
    Port ( i : in STD_LOGIC;
          s : in STD_LOGIC_VECTOR (1 downto 0);
          o : out STD_LOGIC_VECTOR (3 downto 0));
end demux_1_4;

architecture Behavioral of demux_1_4 is
begin
```

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: **0.775 W**
Design Power Budget: **Not Specified**
Process: **typical**
Power Budget Margin: **N/A**
Junction Temperature: **28.7°C**
Thermal Margin: 56.3°C (11.7 W)
Ambient Temperature: 25.0 °C
Effective θ_{JA} : 4.8°C/W
Power supplied to off-chip devices: 0 W
Confidence level: **Low**
[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

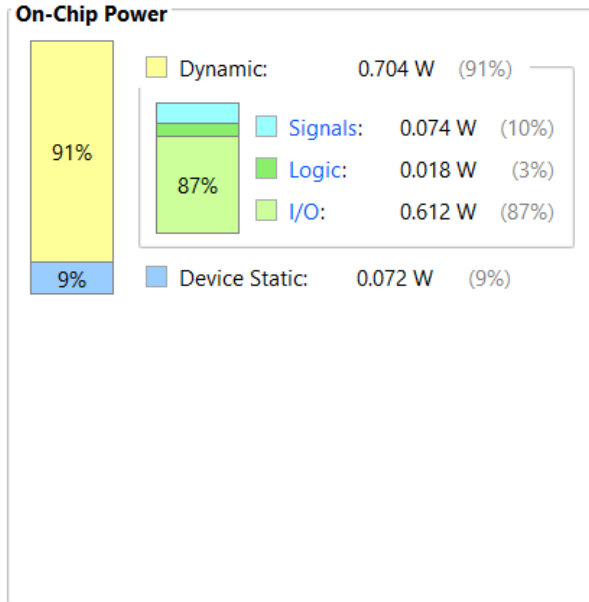


Figure 15: Power Estimation of 16-1 MUX

```
process(s, i)
begin
    o <= "0000";
    case s is
        when "00" => o(0) <= i;
        when "01" => o(1) <= i;
        when "10" => o(2) <= i;
        when "11" => o(3) <= i;
        when others => o <= "0000";
    end case;
end process;

end Behavioral;
```

RTL Schematic Diagram

RTL design is used to describe the behavior of the 1-4 DEMUX as shown in figure 16. It shows the process of how the input and selection lines associate with each other to provide desired output.

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the 1-4 DEMUX circuit as shown in figure 17. It involves converting a VHDL code into a gate-level representation

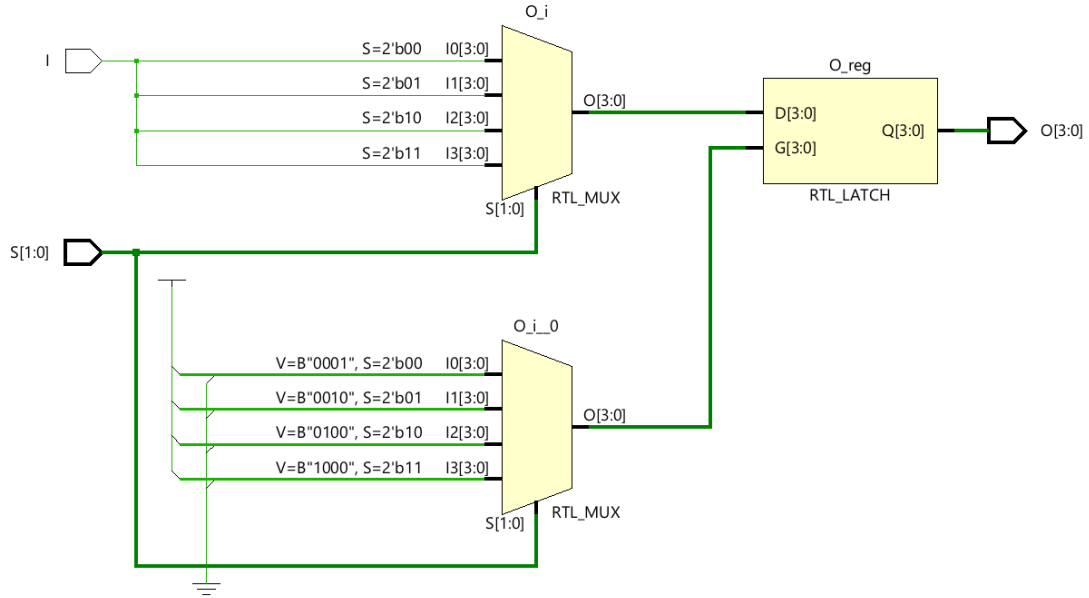


Figure 16: RTL Schematic Diagram of 1-4 DEMUX

through synthesis, generating a netlist that optimizes for factors like speed and power. The resulting schematic visually represents the gates and interconnections, allowing engineers to verify and optimize the circuit's design for efficiency and functionality.

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table of 1-4 DEMUX as shown in figure 18. In the case of a 1-4 DEMUX, the test ensures that the behavior of the VHDL program and the corresponding schematic diagram match the expected behavior as defined by the truth table. The truth table outlines the relationship between the inputs, select lines, and output for all possible combinations.

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 19.

Power Estimation

Figure 20 shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data

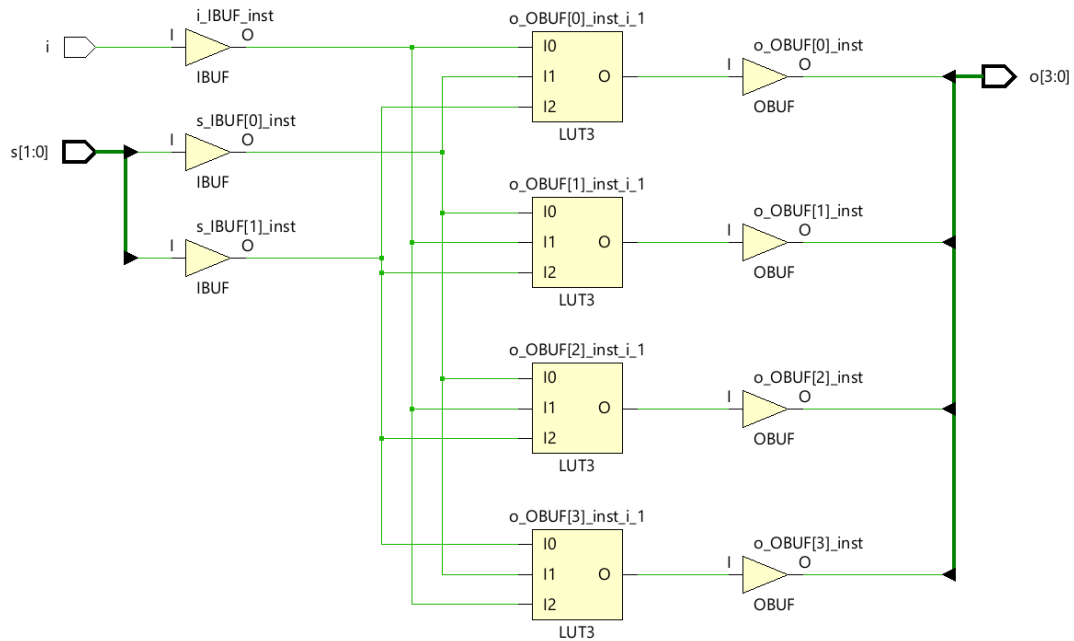


Figure 17: Synthesis Schematic Diagram of 1-4 DEMUX

of a 1-4 DEMUX circuit such as information as junction temperature, thermal margin, confidence level and ambient temperature.

2.5 1-8 DEMUX

VHDL code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity DEMUX1_8 is
    Port ( I : in STD_LOGIC;
          S : in STD_LOGIC_VECTOR (2 downto 0);
          O : out STD_LOGIC_VECTOR (7 downto 0));
end DEMUX1_8;

architecture Behavioral of DEMUX1_8 is

begin
    process(S, I)
    begin
        case S is
            when "000" => O(0) <= I;
            when "001" => O(1) <= I;
            when "010" => O(2) <= I;
```



Figure 18: Timing Diagram of 1-4 DEMUX

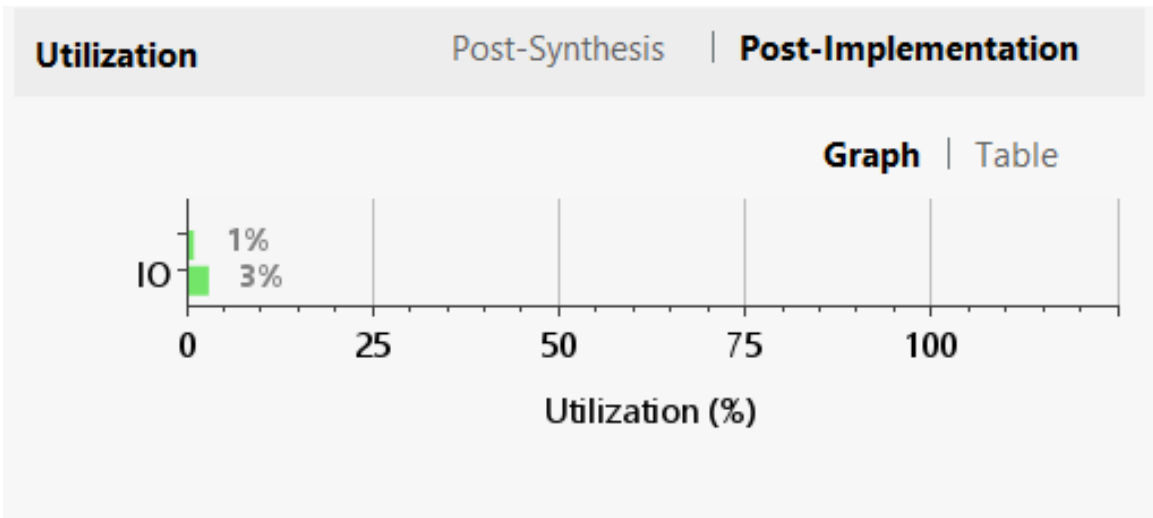


Figure 19: Device Utilization of 1-4 DEMUX

```

when "011" => O(3) <= I;
when "100" => O(4) <= I;
when "101" => O(5) <= I;
when "110" => O(6) <= I;
when "111" => O(7) <= I;
when others => O <= "00000000";
end case;
end process;
end Behavioral;

```

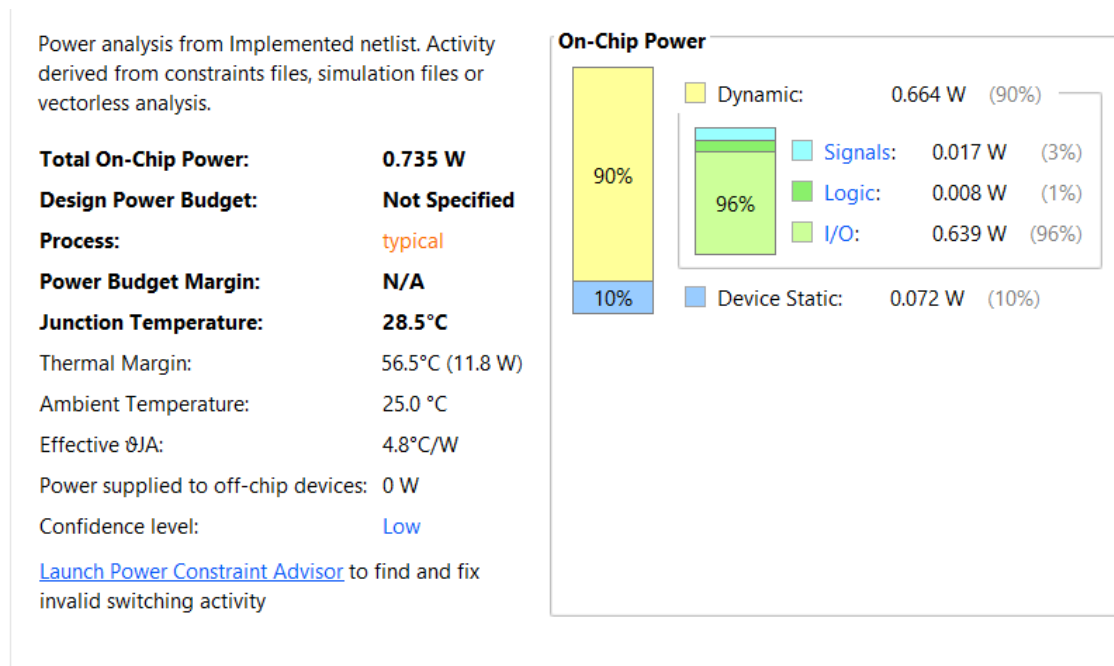


Figure 20: Power Estimation of 1-4 DEMUX

RTL Schematic Diagram

RTL design is used to describe the behavior of the 1-8 DEMUX as shown in figure 21. It shows the process of how the input and selection lines associate with each other to provide desired output.

Synthesis Schematic Diagram

Schematic design is used to describe the implementation of the 1-8 DEMUX circuit as shown in figure 22. It involves converting a VHDL code into a gate-level representation through synthesis, generating a netlist that optimizes for factors like speed and power. The resulting schematic visually represents the gates and interconnections, allowing engineers to verify and optimize the circuit's design for efficiency and functionality.

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table of 1-8 DEMUX as shown in figure 23. In the case of a 1-8 DEMUX, the test ensures that the behavior of the VHDL program and the corresponding schematic diagram match the expected behavior as defined by the truth table. The truth table outlines the relationship between the inputs, select lines, and output for all possible combinations.

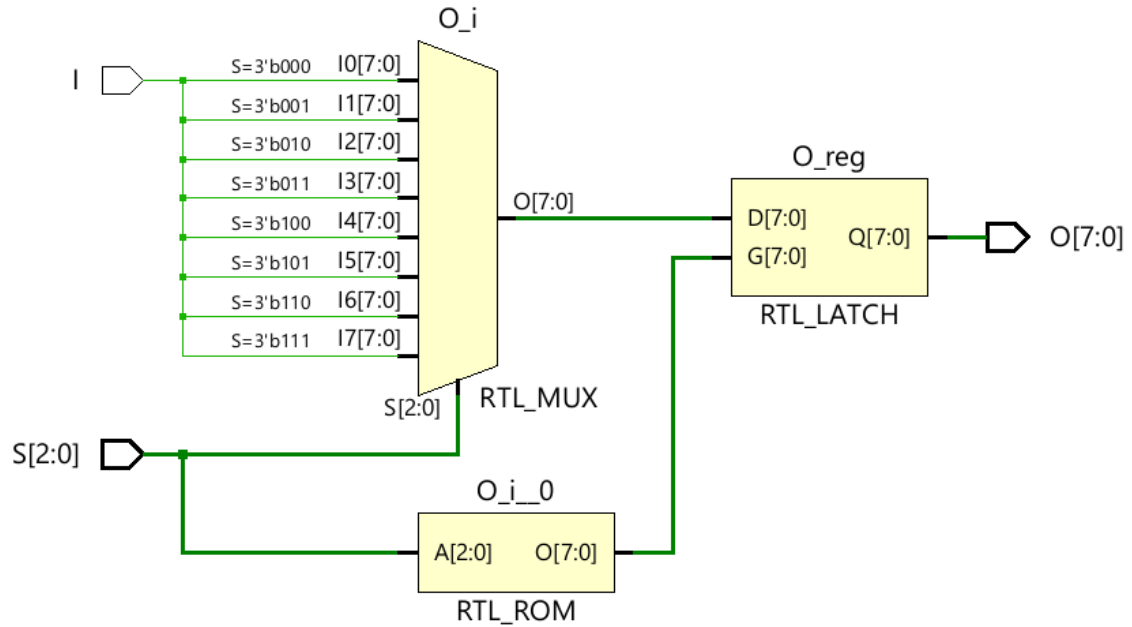


Figure 21: RTL Schematic Diagram of 1-8 DEMUX

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 24.

Power Estimation

Figure 25 shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a 1-8 DEMUX circuit such as information as junction temperature, thermal margin, confidence level and ambient temperature.

2.6 1-16 DEMUX

VHDL code

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity DEMUX1_16 is
    Port ( I : in STD_LOGIC;
          S : in STD_LOGIC_VECTOR (3 downto 0);
```

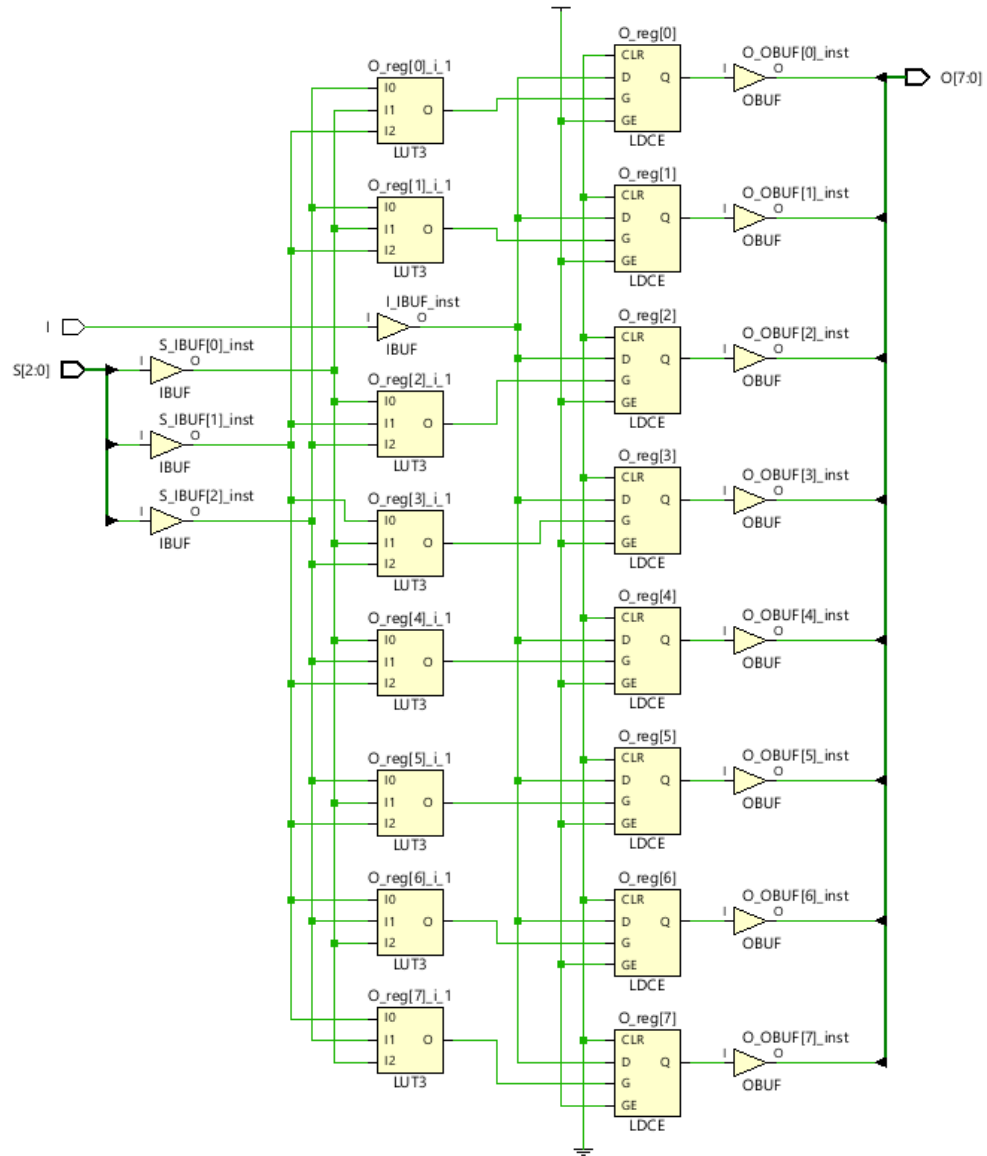


Figure 22: Synthesis Schematic Diagram of 1-8 DEMUX

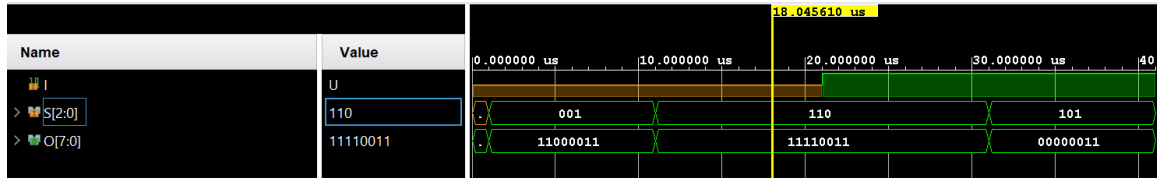


Figure 23: Timing Diagram of 1-8 DEMUX

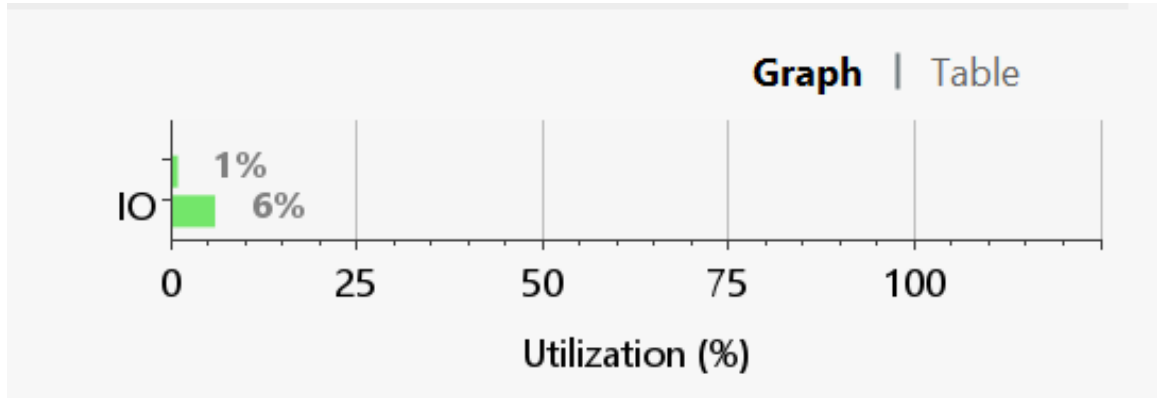


Figure 24: Device Utilization of 1-8 DEMUX

```

        O : out STD_LOGIC_VECTOR (15 downto 0));
end DEMUX1_16;
architecture Behavioral of DEMUX1_16 is
begin
process(S, I)
begin
    case S is
        when "0000" => O(0) <= I;
        when "0001" => O(1) <= I;
        when "0010" => O(2) <= I;
        when "0011" => O(3) <= I;
        when "0100" => O(4) <= I;
        when "0101" => O(5) <= I;
        when "0110" => O(6) <= I;
        when "0111" => O(7) <= I;
        when "1000" => O(8) <= I;
        when "1001" => O(9) <= I;
        when "1010" => O(10) <= I;
        when "1011" => O(11) <= I;
        when "1100" => O(12) <= I;
        when "1101" => O(13) <= I;
        when "1110" => O(14) <= I;
        when "1111" => O(15) <= I;
        when others => O <= "000000000";
    end case;
end process;
end Behavioral;

```

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: **1.821 W**
Design Power Budget: **Not Specified**
Process: **typical**
Power Budget Margin: **N/A**
Junction Temperature: **33.7°C**
Thermal Margin: 51.3°C (10.7 W)
Ambient Temperature: 25.0 °C
Effective θ_{JA} : 4.8°C/W
Power supplied to off-chip devices: 0 W
Confidence level: **Low**
[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

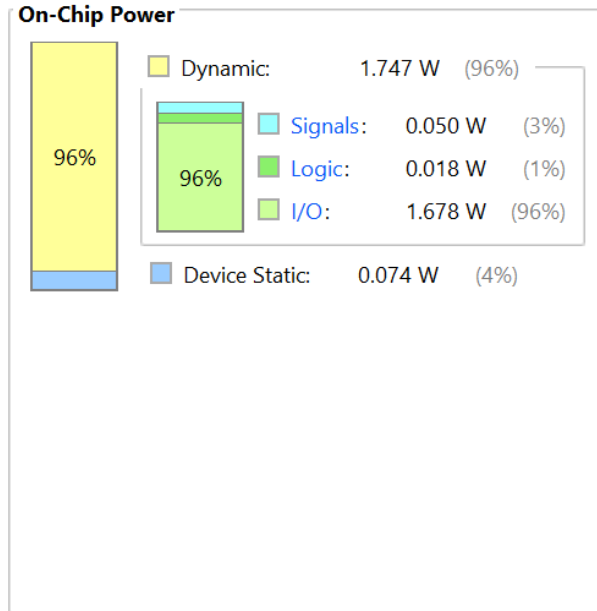


Figure 25: Power Estimation of 1-8 DEMUX

```

end case;
end process;
end Behavioral;

```

RTL Schematic Diagram

RTL design is used to describe the behavior of the 1-16 DEMUX as shown in figure 26. It shows the process of how the input and selection lines associate with each other to provide desired output.

Timing diagram

Timing diagram test is done to check whether the VHDL program and schematic diagram is valid for the given truth table of 1-16 DEMUX as shown in figure 27. In the case of a 1-16 DEMUX, the test ensures that the behavior of the VHDL program and the corresponding schematic diagram match the expected behavior as defined by the truth table. The truth table outlines the relationship between the inputs, select lines, and output for all possible combinations.

Device Utilization

IO (Input/Output) refers to the physical FPGA pins used for communication with external devices, tracking how many of these pins are utilized for inputs and outputs. Utilization, on

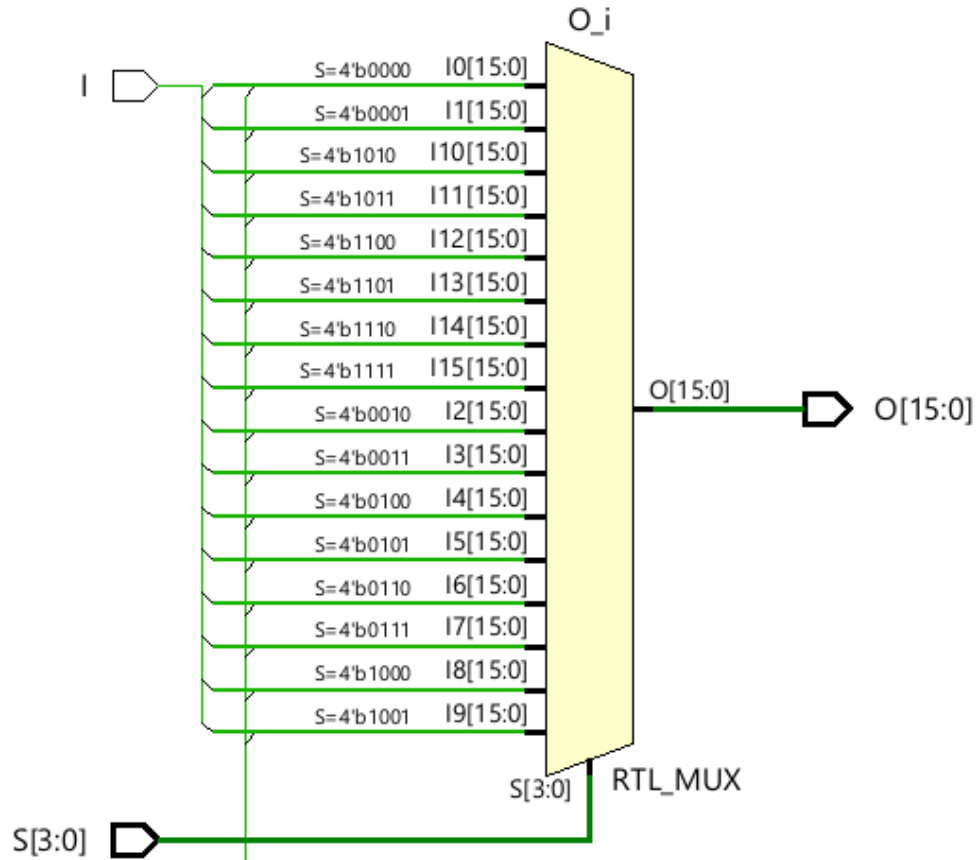


Figure 26: RTL Schematic Diagram of 1-16 DEMUX

the other hand, measures the internal FPGA resource consumption, such as lookup tables (LUTs), flip-flops, block RAM (BRAM), and DSP slices. Together, these concepts provide a comprehensive view of FPGA design efficiency, as shown in figure 28.

Power Estimation

Figure 29 shows estimation of static power (no change in input and output), dynamic power (power required when VHDL program is burned in FPGA) and other related power data of a 1-16 DEMUX circuit such as information as junction temperature, thermal margin, confidence level and ambient temperature.

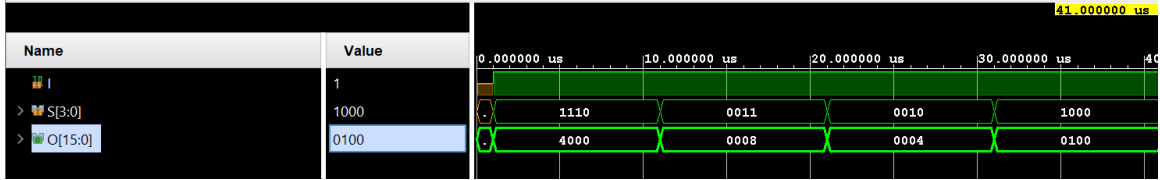


Figure 27: Timing Diagram of 1-16 DEMUX

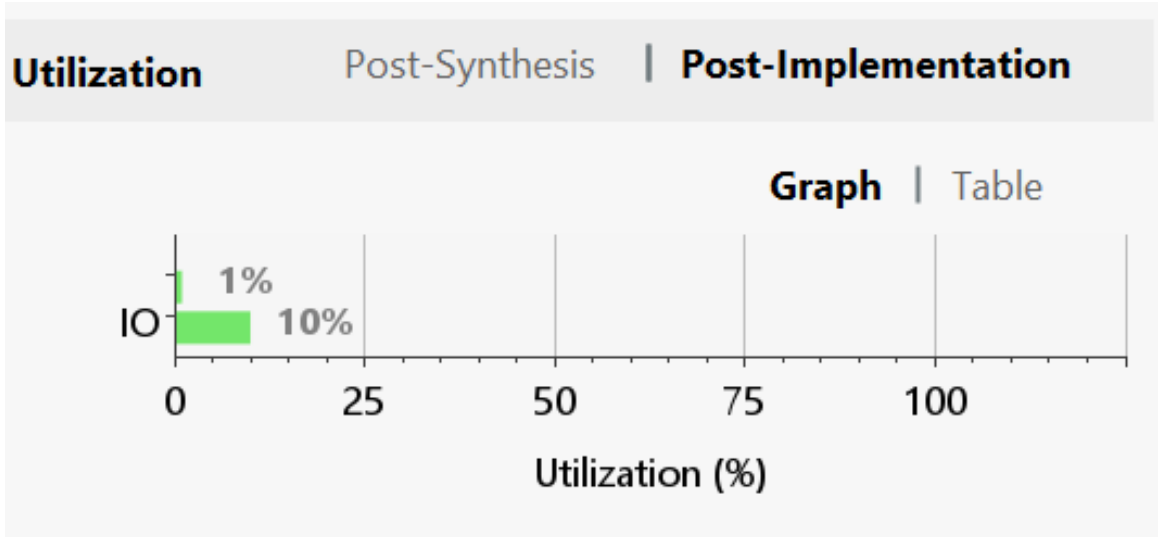


Figure 28: Device Utilization of 1-16 DEMUX

3 Conclusion

In this lab, we implemented the VHDL code for a 4-to-1, 8-to-1, and 16-to-1 VHDL multiplexers and their demultiplexers counterparts on the Atrix-7 FPGA (part: xc7a50tcsg324-1, package: csg324) using Vivado software. Each design was synthesized, simulated, and implemented, followed by hardware testing, which confirmed their correct functionality.

The process began with simulation to verify the timing diagrams, followed by linting to check for syntax errors. Afterward, the designs were synthesized into a hardware logic circuit, and during the implementation phase, power and timing characteristics were analyzed, displayed and studied. A bitstream file was generated, and the necessary pins were assigned for I/O connections. After successful verification, the design was programmed onto the FPGA board and the desired outputs were obtained.

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: **1.821 W**
Design Power Budget: **Not Specified**
Process: *typical*
Power Budget Margin: **N/A**
Junction Temperature: **33.7°C**
Thermal Margin: 51.3°C (10.7 W)
Ambient Temperature: 25.0 °C
Effective θ_{JA} : 4.8°C/W
Power supplied to off-chip devices: 0 W
Confidence level: *Low*

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

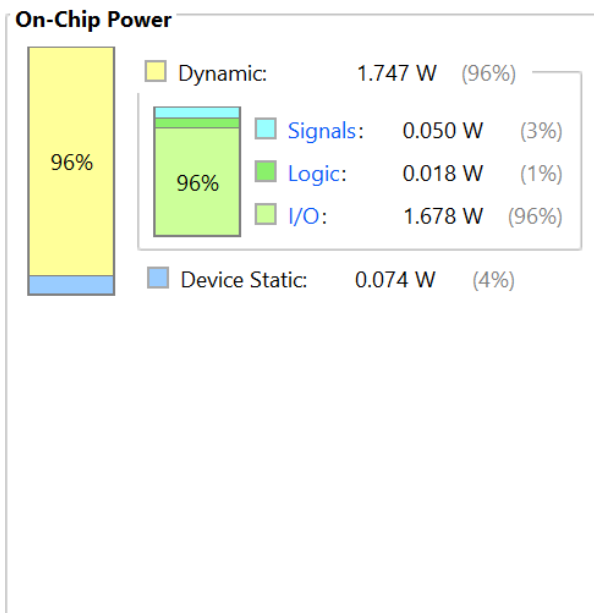


Figure 29: Power Estimation of 1-16 DEMUX